PTEN Regulates PI(3,4)P2 Signaling Downstream of Class I PI3K

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Evidence abstract

The PI3K signaling pathway regulates cell growth and movement and is heavily mutated in cancer.

Class I PI3Ks synthesize the lipid messenger PI(3,4,5)P3.

PI(3,4,5)P3 can be dephosphorylated by 3- or 5-phosphatases, the latter producing PI(3,4)P2.

The PTEN tumor suppressor is thought to function primarily as a PI(3,4,5)P3 3-phosphatase, limiting activation of this pathway.

Here we show that PTEN also functions as a PI(3,4)P2 3-phosphatase, both in vitro and in vivo.

PTEN is a major PI(3,4)P2 phosphatase in Mcf10a cytosol, and loss of PTEN and INPP4B, a known PI(3,4)P2 4-phosphatase, leads to synergistic accumulation of PI(3,4)P2, which correlated with increased invadopodia in epidermal growth factor (EGF)-stimulated cells.

PTEN deletion increased PI(3,4)P2 levels in a mouse model of prostate cancer, and it inversely correlated with PI(3,4)P2 levels across several EGF-stimulated prostate and breast cancer lines.

These results point to a role for PI(3,4)P2 in the phenotype caused by loss-of-function mutations or deletions in PTEN.

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